Observables as Projections of a π-free Graph: A Synthesis of the GeoVac Precision-Observation Arc*

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The Geometric Vacuum (GeoVac) framework separates into two structurally distinct layers, and that separation is the organizing principle of its precision-observation arc. *Layer 1* is a bare combinatorial graph: quantum-number labels $(n, \ell, m, \ldots)$, integer eigenvalues, rational or algebraic matrix elements, no physical units, and—verified symbolically—no π. *Layer 2* is a finite family of named projections from that graph onto continuous physical settings; every physical variable, every dimension, and every transcendental constant that appears in any GeoVac calculation enters through exactly one projection and is traceable to it. This synthesis assembles the four Group 6 papers into a single narrative built on that thesis. The spine is: the projection taxonomy and its three-axis tagging (Paper 34 [10]); time as the prototype *free-side* projection—π enters the Klein–Gordon spectrum on $S^3 \times \mathbb{R}$ at exactly one step, temporal compactification, while rest mass preserves the algebraic ring (Paper 35 [11]); entropy as an embedding-class projection artifact, identically zero on the one-body graph and finite only after the metric embedding of the two-body repulsion (Paper 27 [6]); and the entanglement structure that isolates that repulsion in the angular-momentum eigenbasis (Paper 26 [5]). A multi-observable focal-length program closes the arc: Roothaan autopsies decompose precision atomic observables into projection-chain-tagged components, exposing literature convention mismatches and framework kernel gaps that single-observable analyses cannot see. Two disciplines run throughout. *First*, every transcendental named here is tagged to its projection—π via the Hopf and Fock volume measures, π^{2k} via the spectral action, $\zeta(2k)$ via even-zeta Dirichlet series, odd ζ and Catalan G via the half-integer-Hurwitz spinor lift, 2π via temporal compactification. *Second*, the precision matches are sub-percent *framework-native* agreements that surface convention and kernel issues; they are *not* first-principles predictions of the constants, and the multi-focal-composition wall—the framework has no native theorem for composing multiple Fock-style projections—is named as an active boundary. The unifying reading is that "discreteness is compactness" made operational: the discrete skeleton is the compact/closed regime, and a transcendental is precisely the price of releasing compactness by integrating over a promoted continuous parameter.

I. INTRODUCTION: TWO LAYERS, ONE ORGANIZING OBSERVATION

The GeoVac quantum-chemistry and quantum-computing arcs (Groups 2 and 4) treat the framework as a sparse graph-Hamiltonian engine. The operator-algebra arc (Group 1) treats the same construction as a discrete spectral triple. These two readings are not in conflict; they describe two different *layers* of one object, and the precision-observation arc collected here is the arc that names the layers and the bridge between them.

Layer 1 is the bare combinatorial graph. Its nodes are representation-theoretic labels of compact Lie groups; its Laplacian eigenvalues are integers ($n^2 - 1$ on the Coulomb S^3 graph, N on the Bargmann–Segal S^5 graph); its matrix elements are rational or algebraic over $\mathbb{Q}[\sqrt{2}, \sqrt{3}, \sqrt{6}, \ldots]$ from Wigner $3j$ algebra. That no π appears in Layer 1 is not an aspiration but a verified certificate: on both the Coulomb S^3 Fock graph and the Bargmann–Segal S^5 graph, every adjacency entry, every eigenvalue, and every coupling coefficient is exact rational/algebraic in exact arithmetic (Paper 24 [4], Paper 28's graph-native QED certificate). This is a SYMBOLIC certificate, not a numerical observation.

Layer 2 is a finite family of named maps from the bare graph to physical settings: the Fock 1935 conformal projection, the Hopf bundle, the Bargmann–Segal transform, the Connes–Chamseddine spectral action, the Sturmian reparameterization, the Camporesi–Higuchi spinor lift, the Wigner $3j/D$ couplings, temporal compactification, and others. Each projection introduces specific physical variables (Z, α, $\hbar\omega$, R, β, Λ, $\ldots$), attaches specific dimensions to graph data, and carries a specific transcendental signature. The load-bearing structural claim of the arc is an Observation, asserted at exactly that tier:

Observation 1 (Single-projection origin of physical content)**.** *Every coupling constant, energy scale, length scale, and transcendental constant appearing in any GeoVac computation traces to exactly one projection in Layer 2. The bare graph contains none of α, Z, $\hbar\omega$, R, Λ, π, or $\zeta(2k)$; these enter by, and only by, the projection the calculation invokes.*

This is the operational form of the project's organizing observation that *discreteness is compactness.* Compact groups have discrete spectra (Peter–Weyl), and the bit-exact rational skeleton is the closed/compact regime; the continuum—and with it every transcendental—is what

* Group 6 synthesis in the Geometric Vacuum series

Projection	Dimension	Transcendental signature
Fock conformal	energy	$\kappa = -1/16$; π via $\mathrm{Vol}(S^3)$
Hopf bundle	dimensionless	$\pi = \mathrm{Vol}(S^2)/4$
Bargmann–Segal	energy	*none* (graph π-free)
Spectral action	energy	$\pi^{2k} \cdot \mathbb{Q}$ observables
Spinor lift (CH)	dimensionless	half-int. Hurwitz $\to$ odd ζ; G, $\beta(4)$
Wigner $3j$	dimensionless	$\mathbb{Q}[\sqrt{2k+1}]_k$
Vector-photon	dimensionless	$1/(4\pi)$ per loop
Rest-mass	mass	*trivial* (ring-preserving)
Temporal window	time	$2\pi \cdot \mathbb{Q}$ per Matsubara mode

TABLE I. A slice of the Paper 34 projection dictionary. Each row is a named structural construction that introduces specific physical content into Layer 1 graph data; the transcendental signature is in the sense of Paper 18. The full dictionary catalogues twenty-eight projections.

remains when compactness is released. Calibration data is continuum (un-packed) data, which is why the skeleton is forced and rational while the projections are free and transcendental. The four Group 6 papers make this concrete: Paper 34 catalogues the projections; Paper 35 locates the exact step at which the prototype free-side projection (time) injects π; Papers 27 and 26 show entropy and entanglement to be projection artifacts, zero on the bare graph and finite only after metric embedding.

We hold one framing discipline throughout, following the GeoVac rhetoric rule: the discrete graph and its continuous projections are dual descriptions connected by proven maps, and no ontological priority is asserted for either. The contribution is a vocabulary that makes the question "where does this π come from?" answerable, not a claim about which description is fundamental.

II. THE PROJECTION TAXONOMY AND MASTER-MELLIN TAGGING (PAPER 34)

Paper 34 is the keystone of this group: the living catalogue of *twenty-eight* projection mechanisms, each tagged along three independent axes—(a) the physical variables it introduces, (b) the dimension or unit it attaches to graph data, and (c) its transcendental signature in the sense of Paper 18 [3]. The transcendental axis is a structural sibling of Paper 18's transcendental classification: each tier corresponds to a projection type, and reading the dictionary in either direction recovers the other. A representative slice of the dictionary is reproduced in Table I.

a. Where the transcendentals live. The dictionary makes the tagging discipline mechanical. The kinetic constant $\kappa = -1/16$ and the volume π enter at the Fock conformal projection; the dimensionless $\pi = \mathrm{Vol}(S^2)/4$ at the Hopf bundle; the $\pi^{2k} \cdot \mathbb{Q}$ content of vacuum-polarization-type observables (e.g. $\Pi = 1/(48\pi^2)$) at the Connes–Chamseddine spectral action; the even-zeta value $\zeta_R(2)$ (the Fock-Dirichlet object F) at a Dirichlet series at $s = 4$; the half-integer Hurwitz content that promotes to odd ζ, Catalan's G, and $\beta(4)$ at the Camporesi–Higuchi spinor lift's vertex selection; and $2\pi \cdot \mathbb{Q}$ at temporal compactification (Sec. III). None of these is an anonymous transcendental: each is pinned to a projection row, which is the discipline this group exists to model. The fine-structure combination $1/(4\pi)$ per loop is tagged to the vector-photon promotion; the Paper 2 fine-structure combination rule that consumes it remains an **Observation**, never a derivation, and the taxonomy is consistent with that tiering rather than strengthening it.

b. The engine versus the catalogue. The computational engine behind the transcendental tags is the master Mellin engine—the Mellin transform of $\mathrm{Tr}\,(D^k e^{-tD^2})$ at $k \in \{0, 1, 2\}$, whose three domains $M_1/M_2/M_3$ generate the π^{2k}, even-zeta, and odd-zeta/Catalan tiers respectively. That engine is developed in Paper 18 §III.7 (Group 3, the foundations trunk); Paper 34 is the *catalogue* that consumes its output and assigns each tier to a projection. The synthesis-level reading is that the transcendental axis is the only genuinely independent axis of the three: Sprint TX-A's[**?**] dimensional axiomatization found the variable and dimension axes to be perfectly correlated, so the dictionary's content reduces to the projection $\leftrightarrow$ transcendental map.

III. TIME AS THE PROTOTYPE FREE-SIDE PROJECTION (PAPER 35)

Paper 35 answers the question Paper 34 leaves open—*where exactly* does π enter, and at no other step—in the simplest non-trivial relativistic system the framework describes: a free scalar on $S^3 \times \mathbb{R}$ governed by the Klein–Gordon equation.

a. The before/after. The bare Klein–Gordon spectrum is π-free in the algebraic-extension ring $\mathbb{Q}[\sqrt{d_i}]$ for every rational mass-squared parameter. This is verified SYMBOLICALLY for $n \in [1, 50]$ across the panel $m^2 \in \{0, 1, 1/4, 2\}$ (200 cases, zero transcendentals). Mass is therefore a *rest-mass projection*: it introduces a parameter of dimension mass and *preserves* the bare-graph ring. Compactify the temporal direction, $S^3 \times \mathbb{R} \to S^3 \times S^1_\beta$, and the spectrum acquires Matsubara modes $\omega^t_k = 2\pi k/\beta$; the first π-bearing eigenvalue is the (n=0, k=1) mode with $\omega^2 = 4\pi^2$. Every $(n, 0)$ eigenvalue is π-free; every $(n, k \neq 0)$ eigenvalue carries $2\pi/\beta$. The conformally coupled massless scalar Casimir energy on unit S^3 (spatial only, no temporal compactification) is the exact rational $E_{\mathrm{Cas}} = 1/240$, with no transcendental at any step; the Stefan–Boltzmann constant $\pi^2/90$ appears only in the high-temperature expansion on $S^3 \times S^1_\beta$—through the Matsubara sum, i.e. exactly the spectral π-injection.

b. The sharpened reading and its tier. This forces a refinement of the Paper 34 dictionary: the implicit "scalar parameter projection" splits into a ring-preserving *rest-mass projection* (introduces m, signature trivial) and an *observation/temporal-window projection* (introduces β, signature $2\pi \cdot \mathbb{Q}$ per mode). The structural statement is that a GeoVac observable contains π if and only if its evaluation includes a continuous integration over a temporal or spectral parameter promoted from the discrete graph spectrum. This is a structural OBSERVATION supported by a symbolic verification panel (the 208-case panel that Paper 34's prediction subsection cites as a downstream control); it is asserted at that tier, not as a theorem about all observables. Time is, in this reading, the prototype free-side projection: the observer's finite-time measurement window is the formal mechanism by which the continuum—and π—enters.

This connects directly to the project's internal working hypothesis that time-discreteness is observer-compactification (WH7): the only temporal structure the metric can see is compactified time, and compactified time is automatically discrete. We flag the honest cap that internal diagnostic work attaches to that hypothesis—whether the Lorentzian *signature* is metrically visible at finite cutoff remains open, and the relevant falsifier (a metrically visible temporal algebra on a non-compact carrier) is an active probe, not a closed result. Paper 35's load-bearing content—that π enters at temporal compactification and that rest mass is ring-preserving—stands independently of that open question.

IV. ENTROPY AS AN EMBEDDING-CLASS PROJECTION (PAPER 27)

Paper 27 places entanglement entropy in the same two-layer frame and shows it to be a projection artifact rather than an intrinsic feature of the graph.

a. One-body operators are entanglement-inert. The operator-theoretic result is an INTERNAL THEOREM: on any finite lattice, the von Neumann entropy of any reduced density matrix derived from a *non-degenerate* ground state of a purely one-body fermionic Hamiltonian is identically zero (the ground state is a single Slater determinant). Numerically this is confirmed to machine precision, $S_{\mathrm{kin}}/S_{\mathrm{full}} \sim 10^{-14}$ for He at $n_{\max} = 2, 3$. All ground-state entanglement is generated by the two-body interaction V_{ee}. The non-degeneracy qualifier is essential—open-shell and dissociated-bond states escape the floor—and is stated as such.

b. The "area law" is pair counting. Earlier framework material presented a holographic scaling $S_n = k \log A_n$ with $A_n \propto n^4$ in one-particle language. Paper 27 re-derives it rigorously as a *pair-counting* statement: $A_n = g_n^2$ with shell degeneracy $g_n = 2n^2$, so the factor of four is the two-body signature, not a genuine area law. This corrects the earlier presentation in place rather than propagating it—a model of the two-way verdict discipline (a claim is re-priced to what its backing supports).

c. A second rigidity statement. On the Bargmann–Segal harmonic-oscillator lattice, the two-fermion closed-shell ground-state entropy is identically zero for *any* central two-body potential. The mechanism is an INTERNAL THEOREM: Moshinsky–Talmi total-quanta conservation makes $N_1 + N_2 = N_{\mathrm{rel}} + N_{CM}$ a good quantum number, so the closed-shell ground state is the one-dimensional lowest-quanta sector—a single determinant $|(0s)^2\rangle$, confirmed bit-identically at $N_{\max} = 2, 3$ with commutator residual $\|[H_{HO}, V]\|_F/\|H_{HO}\|_F \sim 10^{-16}$. This is the entropy-side dual of the Fock and Bargmann–Segal rigidity theorems of Paper 24.

d. Taxonomy placement. Entropy slots cleanly as an *embedding-class* observable in the Paper 18 taxonomy and a projection artifact in the Paper 34 frame: zero on the combinatorial graph (one-body operator alone, no transcendental), finite only after the metric embedding of $1/r_{12}$ onto the pair-state graph, where the spectral eigenvalues of V_{ee} become irrational. The combinatorial graph is entropy-blind; entropy is the price of projecting onto the metric continuum. A measured refinement accompanies the placement: Coulomb-lattice block entropy obeys a universal two-variable scaling $S_B = A\,(\tilde{w}_B/\delta_B)^{\gamma}$ whose local exponent Richardson-extrapolates to $\gamma_\infty \approx 1.96$, slightly below the non-degenerate second-order Rayleigh–Schrödinger value of 2—a MEASURED extrapolation, reported with its residual gap attributed to multi-shell aggregation, not claimed as exactly 2.

V. ENTANGLEMENT STRUCTURE OF THE ANGULAR-MOMENTUM BASIS (PAPER 26)

Paper 26 supplies the computational backbone for the entropy thesis by characterizing *where* the two-body entanglement lives in the angular-momentum eigenbasis—the basis of hydrogenic radial functions times spherical harmonics labeled by (n, ℓ, m).

a. Energy–entanglement decoupling. The off-diagonal one-body Hamiltonian (graph-Laplacian topology, coupling $\kappa = -1/16$) carries 10–39% of the correlation energy but $< 0.2\%$ of the single-orbital entanglement entropy; the off-diagonal electron repulsion carries 100% of the entanglement. The total entanglement entropy scales as $S \sim Z^{-2.56}$ across He-like ions. These are MEASURED quantities across $Z = 2$–10 at $n_{\max} = 2$–4. The structural origin is that a one-body operator is separable and cannot entangle orbitals, while V_{ee} is the irreducible two-body interaction; the angular-momentum basis, by nearly diagonalizing the one-body physics, isolates the entanglement in V_{ee} alone.

b. Basis-intrinsic sparsity. Among orbital bases for spherically symmetric systems, the angular-momentum eigenbasis is an *isolated* point of ERI sparsity: any unitary rotation of orbital indices, however small, fills the ERI tensor from 42% to 99% density in a step-function transition at the identity, and the nonzero ERI count is Z-independent (sparsity is purely geometric). This MEASURED step-function is the entanglement-side counterpart of the potential-independent angular-sparsity theorem (Paper 22, Group 3) and the structural root of the $O(Q^{2.5})$ composed Pauli scaling reported in the quantum-computing arc (Group 4): the same basis that decouples one-body energy from two-body entanglement is the one that produces sparse ERIs.

c. Network topology and the factorization it justifies. Orbital mutual-information maps exhibit a hub-migration pattern ($1s \to 2s \to 2p$) tracking the partially filled shell across the first row, and core–valence mutual information falls to $\approx 2 \times 10^{-3}$ by $Z = 4$ and below 10^{-3} for $Z \geq 5$. The latter is the quantitative justification for the composed fiber-bundle factorization used in the molecular framework: per-block entanglement is R-independent, with core/bond entropy ratios of $\sim 50\times$ at matched qubit count. Paper 26 states its limitations plainly—$n_{\max} = 2$ for the first-row network maps, R-independence is a property of the composed architecture (not of the true molecular ground state, whose entanglement diverges at dissociation), and the d-orbital extension is open.

VI. THE MULTI-OBSERVABLE FOCAL-LENGTH PROGRAM (PAPER 34 §V)

The arc's most distinctive instrument is the multi-observable focal-length decomposition program: the discovery that a global fit of several precision observables through one structural dictionary exposes systematics that no single-observable analysis can see.

a. Roothaan autopsies. For observables with a multi-component literature decomposition—the hydrogen Lamb shift, the 21 cm and muonic hyperfine intervals, the muonic-deuterium Lamb shift—Paper 34 hosts *Roothaan autopsies*: decomposition tables that present each measured observable as a sum of focal-length-tagged components, each carrying its explicit projection chain and a framework-native (FN) versus Layer-2-input (L2) status. The hydrogen $1S$ Lamb-shift autopsy is representative: the framework-native components (self-energy via Fock $\circ$ Sturmian $\circ$ spectral action $\circ$ Drake–Swainson, Uehling vacuum polarization via Fock $\circ$ spectral action, ...) sum to +1057.19 MHz, 99.9% of the 1057.845(9) MHz measurement, with the Layer-2 inputs (multi-loop QED, finite-size, recoil NLO) summing to a near-cancellation specific to this observable. These are MEASURED decompositions whose residuals sit within the cited compilation precision; the autopsy makes the projection-chain dictionary's reach visible at the observable level.

b. Convention exposures versus kernel gaps. The program's signature finding-type is the *literature convention exposure*: a multi-observable global fit surfaces itemization differences between independent precision compilations (Eides, Karshenboim, Krauth, Pachucki–Yerokhin, Antognini, Friar–Payne) at the sub-percent level the framework operates at. Six are catalogued, including: the $\sim 0.01\%$-of-ν_F recoil/multi-loop itemization difference that propagates to ~ 25 mfm in an extracted Zemach radius; the Friar-moment convention difference between an RMS radius $\langle r^2\rangle^{1/2}$ and a first moment $\langle r\rangle$ (a Gaussian-profile factor $\langle r^2\rangle/\langle r\rangle^2 = 4/\pi \approx 1.27$, i.e. +12.7% on the Friar moment); and the resolution of a 3σ helium fine-structure disagreement traced to a numerically incorrect relativistic Bethe-log itemization in an older compilation. The complementary finding-type is the *framework kernel gap*: places where the framework's leading-order operator (Eides Zemach, Foldy–Friar contact, Friar moment) is too coarse for the precision target. Each precision tightening tends to relocate the load-bearing systematic from one to the other; cataloguing which class a given observable lands in is itself the program's output.

c. The structural prediction that replaced the naive one. A falsifiable structural PREDICTION emerges from re-tagging the matches catalogue by the number of Layer-2 inputs a chain consumes. An earlier form scaled the residual with projection *depth*; the May 2026 synthesis pass falsified that form on the catalogue itself (depth-3 chains exist at residual 0 and at +286 ppm; depth-4 chains at +2 ppm and at +0.534%). The replacement is the *Layer-2-presence bound*: the residual of a catalogue row is bounded above by the larger of the framework's basis-quality residual at zero Layer-2 inputs and the largest single Layer-2 input consumed,

$$|\epsilon(\mathrm{obs})| \;\leq\; \max\!\big(\epsilon_{\mathrm{basis}},\; \max_i |L_2^{(i)}|\big).$$

Zero-Layer-2 chains run from machine-exact (Stefan–Boltzmann $\pi^2/90$, hydrogen static polarizability $\frac{9}{2}a_0^3$, S^3 Casimir 1/240) to basis-quality ceilings (-0.534% for the one-loop Lamb-shift closure); one-input chains tighten to tens of ppm or below; multi-input chains sit in the multi-focal-wall regime, dominated by the largest single input. This is a PREDICTION, consistent with the present catalogue and falsifiable by future calculation, not a phenomenological fit.

VII. HONEST SCOPE AND POSITIONING

a. What is robust. The load-bearing results are structural and verified at their stated tiers. The π-free Layer 1 certificate is SYMBOLIC (exact arithmetic, both Coulomb S^3 and Bargmann–Segal S^5). The Klein–Gordon π-injection at temporal compactification is a SYMBOLIC panel (200 cases; $E_{\mathrm{Cas}}^{S^3} = 1/240$ and the conformal-scalar chain exact). The one-body entanglement-inertness and the HO zero-entropy rigidity are INTERNAL THEOREMS with machine-precision numerical confirmation. The two-layer thesis and the single-projection origin of physical content are stated as

OBSERVATIONS, asserted at that tier and no higher. The projection dictionary is a concrete, structured artifact: every transcendental in the arc wears its projection tag.

b. What is conditional. The precision matches are sub-percent *framework-native* agreements, not first-principles predictions of the physical constants. Each consumes external Layer-2 inputs (Zemach radii, g-factors, multi-loop QED coefficients, nuclear polarizabilities) in their native compilation conventions, and the program's value is precisely that it surfaces where those conventions and the framework's leading-order kernels are too coarse—a diagnostic instrument for precision systematics, not a replacement for the precision compilations. The block-entropy exponent $\gamma_\infty \approx 1.96$ is a MEASURED extrapolation, not the exact second-order value 2. The entropy area-law and the R-independence of composed block entanglement are properties of the construction, stated with their limitations, not physical statements about molecular entanglement at dissociation.

c. The active wall. The arc's principal open boundary is the *multi-focal-composition wall*: the framework couples discrete labels cleanly but has *no* native composition theorem for combining multiple Fock-style projections. Six independent instances are on record across domains (the Yukawa non-selection theorem, the LS-8a renormalization scale, the HF spatial recoil/Zemach/multi-loop attempts, the chemistry W1e wall). This is why multi-Layer-2-input observables sit in the residual-dominated regime of the Layer-2-presence bound, and why the precision program's deepest residuals are structural rather than numerical. The wall is named as an active boundary, not papered over.

d. Positioning. The natural reading of this arc is as a *vocabulary and a diagnostic*. The two-layer framework gives every GeoVac observable a definite answer to "where does each transcendental enter," and the multi-observable focal-length program turns that vocabulary into an instrument that exposes convention mismatches and kernel gaps in precision atomic physics. The discrete skeleton and its continuous projections are dual descriptions; the contribution is structural traceability, not an ontological claim and not a derivation of the constants the projections consume.

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[1] J. Loutey, "Angular Momentum as Information Geometry: Isotropic Packing and the Universal Angular Cross-Section," GeoVac Paper 0 (2026).

[2] J. Loutey, "The Dimensionless Vacuum: Recovering the Schrödinger Equation from Scale-Invariant Graph Topology," GeoVac Paper 7 (2026).

[3] J. Loutey, "Spectral-Geometric Exchange Constants: Projecting Discrete Spectra onto Continuous Manifolds," GeoVac Paper 18 (2026). [Master Mellin engine, §III.7; transcendental-tier classification.]

[4] J. Loutey, "The Bargmann–Segal Lattice: π-Free Discretization of the Harmonic Oscillator on S^5," GeoVac Paper 24 (2026). [Layer-1 π-free certificate; HO rigidity theorem.]

[5] J. Loutey, "Entanglement Structure of the Angular Momentum Eigenbasis: Energy-Entanglement Decoupling and Basis-Intrinsic Sparsity," GeoVac Paper 26 (2026).

[6] J. Loutey, "Entropy as a Projection Artifact: One-Body Operators are Entanglement-Inert on Sparse Lattices," GeoVac Paper 27 (2026).

[7] J. Loutey, "QED on S^3: Spectral Geometry of Perturbative Transcendentals," GeoVac Paper 28 (2026). [Graph-native QED π-free certificate.]

[8] J. Loutey, "The Universal/Coulomb Partition of the GeoVac Spectral Triple," GeoVac Paper 31 (2026).

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[10] J. Loutey, "GeoVac as a Two-Layer Framework: Projection Taxonomy and the Empirical Matches Catalog," GeoVac Paper 34 (2026).

[11] J. Loutey, "Time as Projection: The Klein–Gordon Spectrum on $S^3 \times \mathbb{R}$ and the Algebraic / Observation Split," GeoVac Paper 35 (2026).